UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Flowcharts and visuals

Procedures, decision ladders and comparison tables, built from the Flowchart Template blocks.

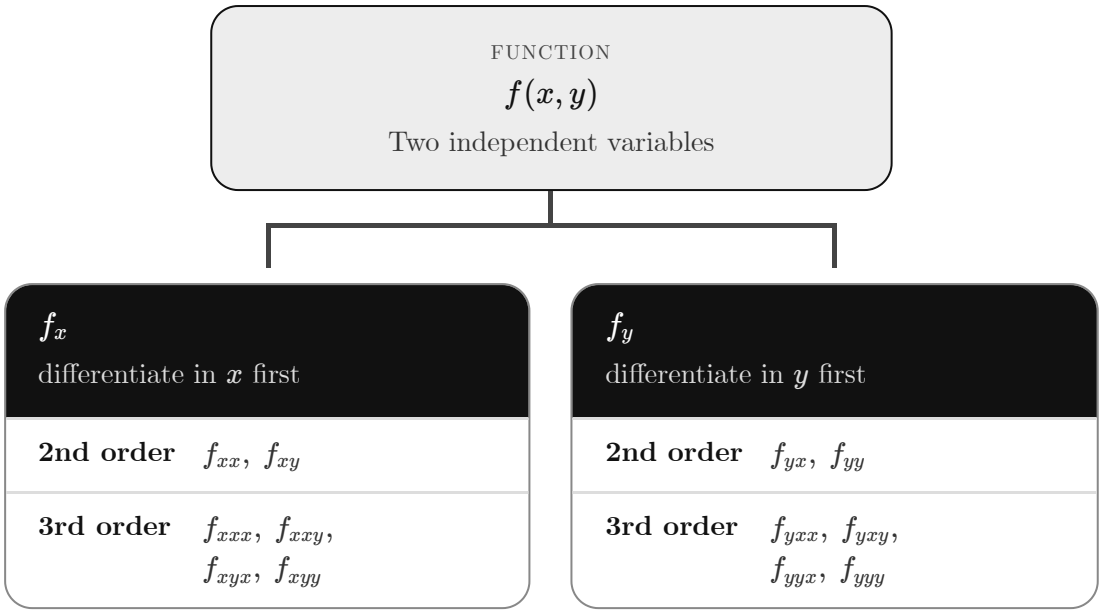
Class 002 · Geometrical interpretation of partial derivatives · compiled 14 October 2026

ESA ★★	Computed in past ESA papers or class-work problems
ESA ★	Conceptual: explained in class, not examined so far
ADDED FOR CLARITY	Standard context that is not on the class slides
TRANSCRIPT ONLY	Said in the lecture but not on any slide
S002·7	Deck 002, slide 7 (rendered .pptx)
T002 00:06:10	AV-summary transcript 002, segment start time
V002 00:07:45	Frame grabbed from the class 002 video
CW U1 P2 · ESA Dec-23 1a	Unit 1 class-work problem 2 · past paper question

Contents

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5. Figure F4 · Reading concavity
6. Figure F5 · Verifying equal mixed partials
7. Figure F7 · Higher-order partials at a point
8. One variable vs two variables

1 Higher-order partial derivatives

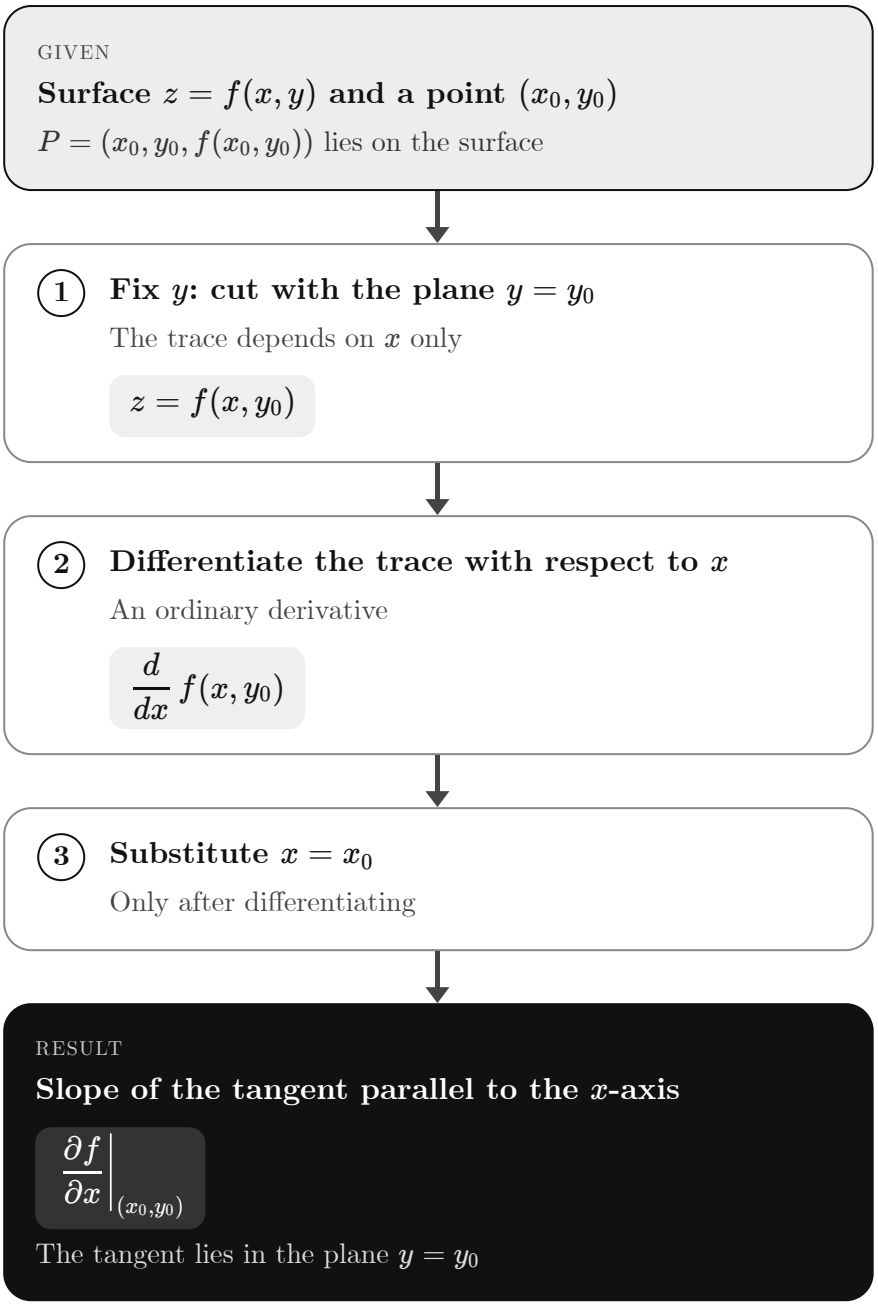


Remark. Totals: 2, 4, 8, ..., 2^n. With m variables there are m^n. When mixed partials are equal, only $n+1$ of the 2^n are different.

Figure F6. Counting partial derivatives of each order. ESA ★

S002·3 · T002 00:01:37 · T002 00:02:13

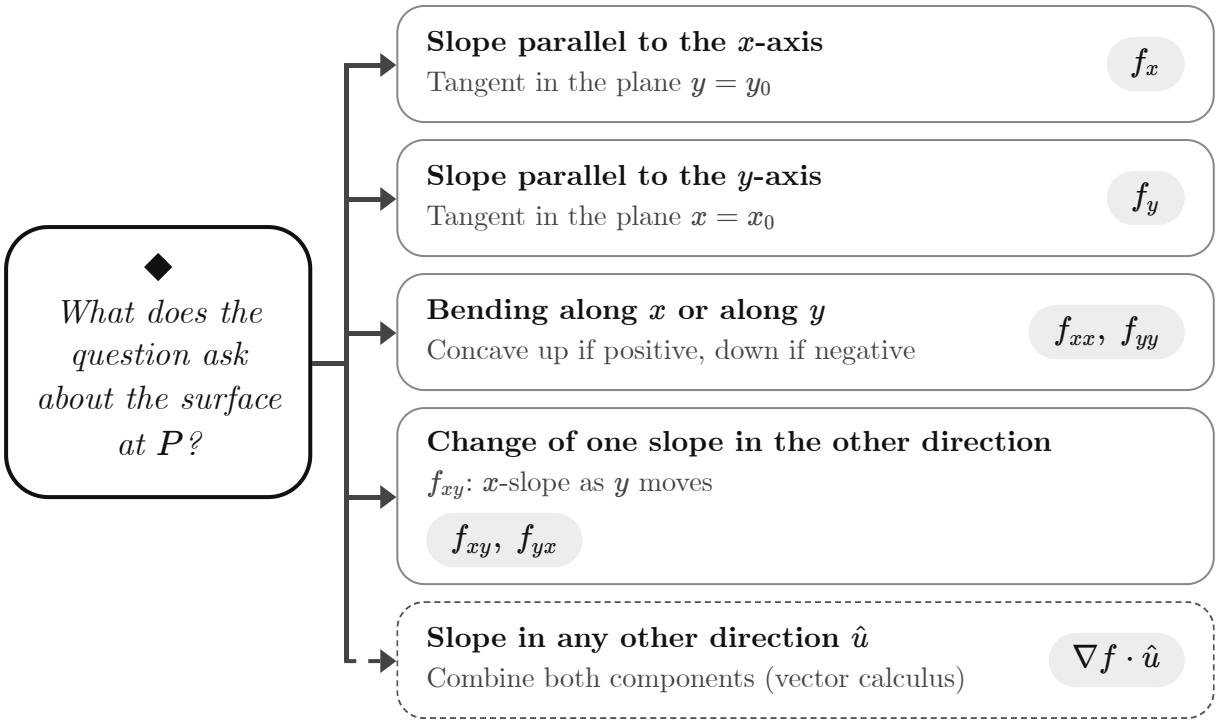
4 First-order partials as slopes of tangents



Remark. For $\partial f/\partial y$, swap the roles: cut with $x = x_0$, differentiate $f(x_0, y)$ in y, then put $y = y_0$.

Figure F1. Partial derivative as the slope of a tangent, step by step. ESA ★★

S002·6 · S002·7 · V002 00:07:45 · T002 00:06:10

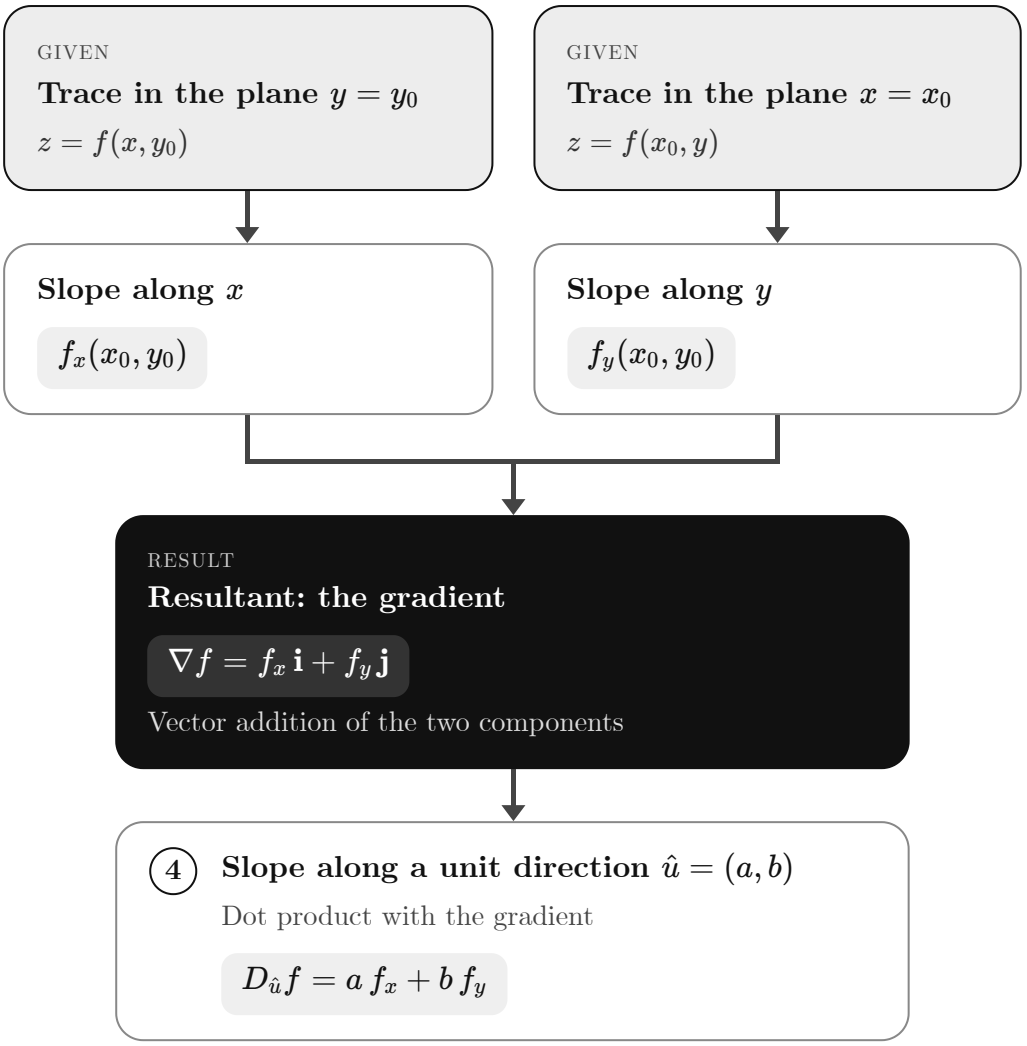


Remark. The dashed route is the gradient idea from slide 8. Its formula $\nabla f \cdot \hat{u}$ is added here for clarity and studied in vector calculus.

Figure F2. Which derivative answers which geometric question. ESA ★★

ADDED FOR CLARITY S002·6 · S002·8 · S002·9 · T002 00:07:54

5 Slope in any other direction: the gradient

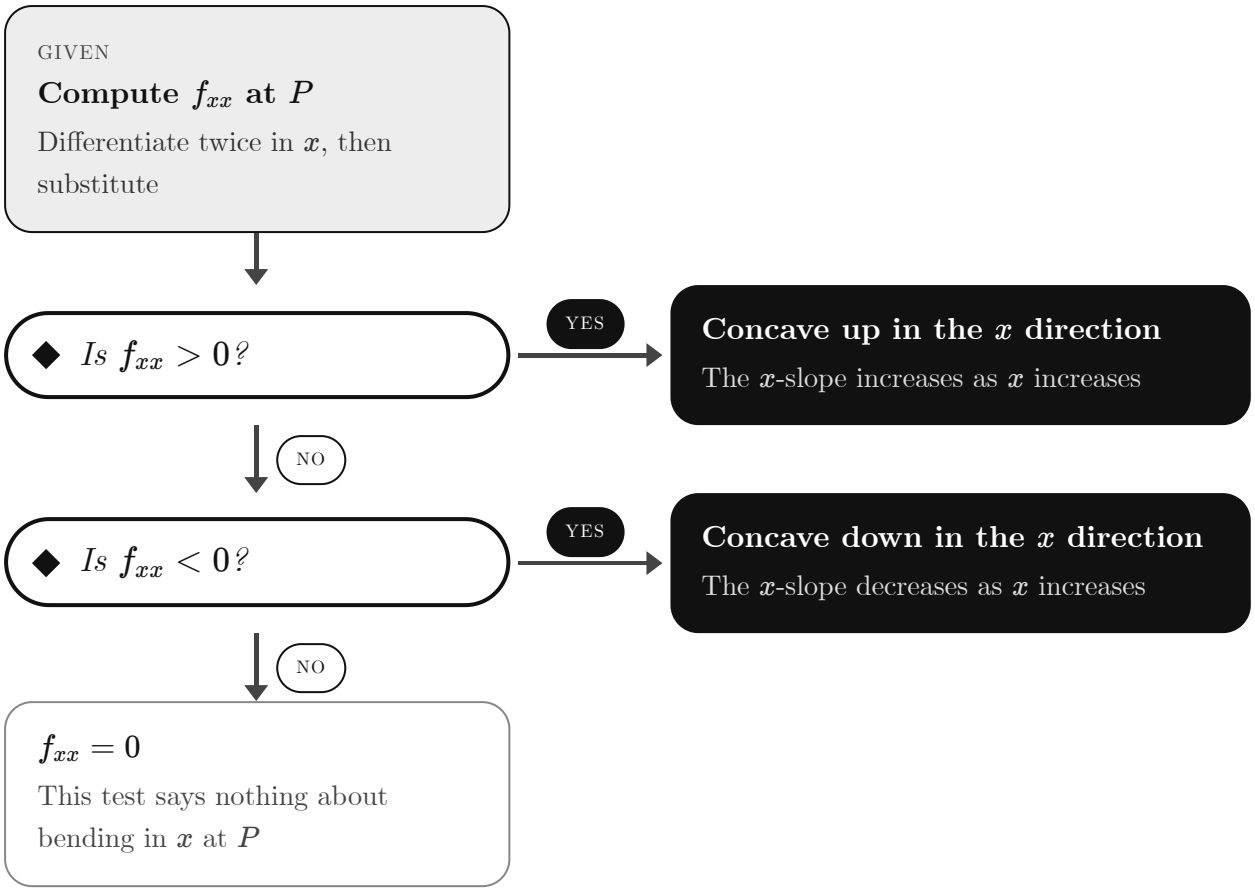


Remark. Check $\hat{u} = (1, 0)$: you get back f_x. The steepest slope is $|\nabla f| = \sqrt{f_x^2 + f_y^2}$. Example: the dome at $(1, -0.5)$ gives $\nabla f = (-2, 1)$ (W2).

Figure F3. Two slopes combine into the gradient. ESA ★

ADDED FOR CLARITY S002·8 · T002 00:09:06 · T002 00:09:23

6 Second-order partials: concavity and mixed partials

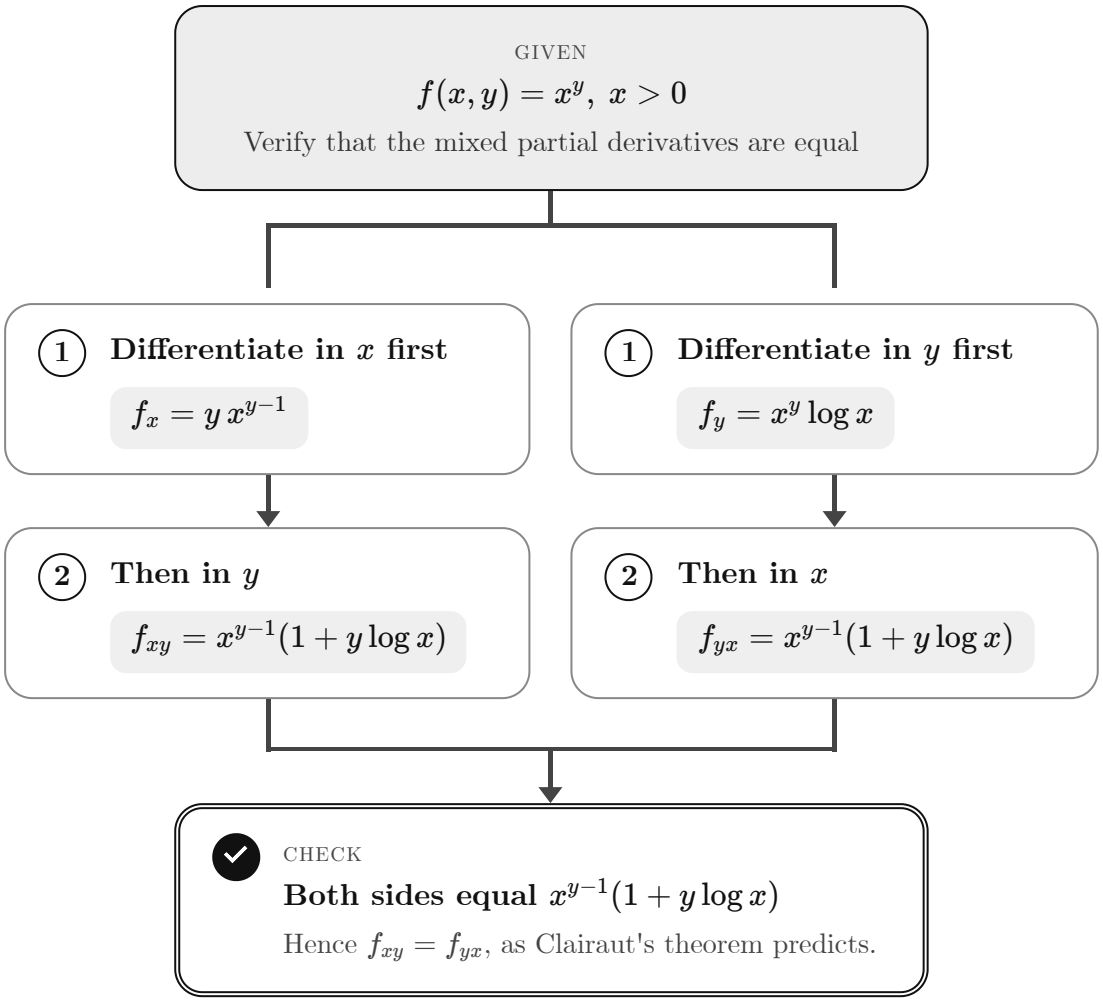


Remark. Run the same ladder with f_{yy} for the y direction. Read the sign of f_{xy} in the same way: positive means the x-slope increases as y increases (Figure G10).

Figure F4. Reading concavity from a second-order partial. ESA ★★

ADDED FOR CLARITY S002·9 · T002 00:11:19 · T002 00:11:38

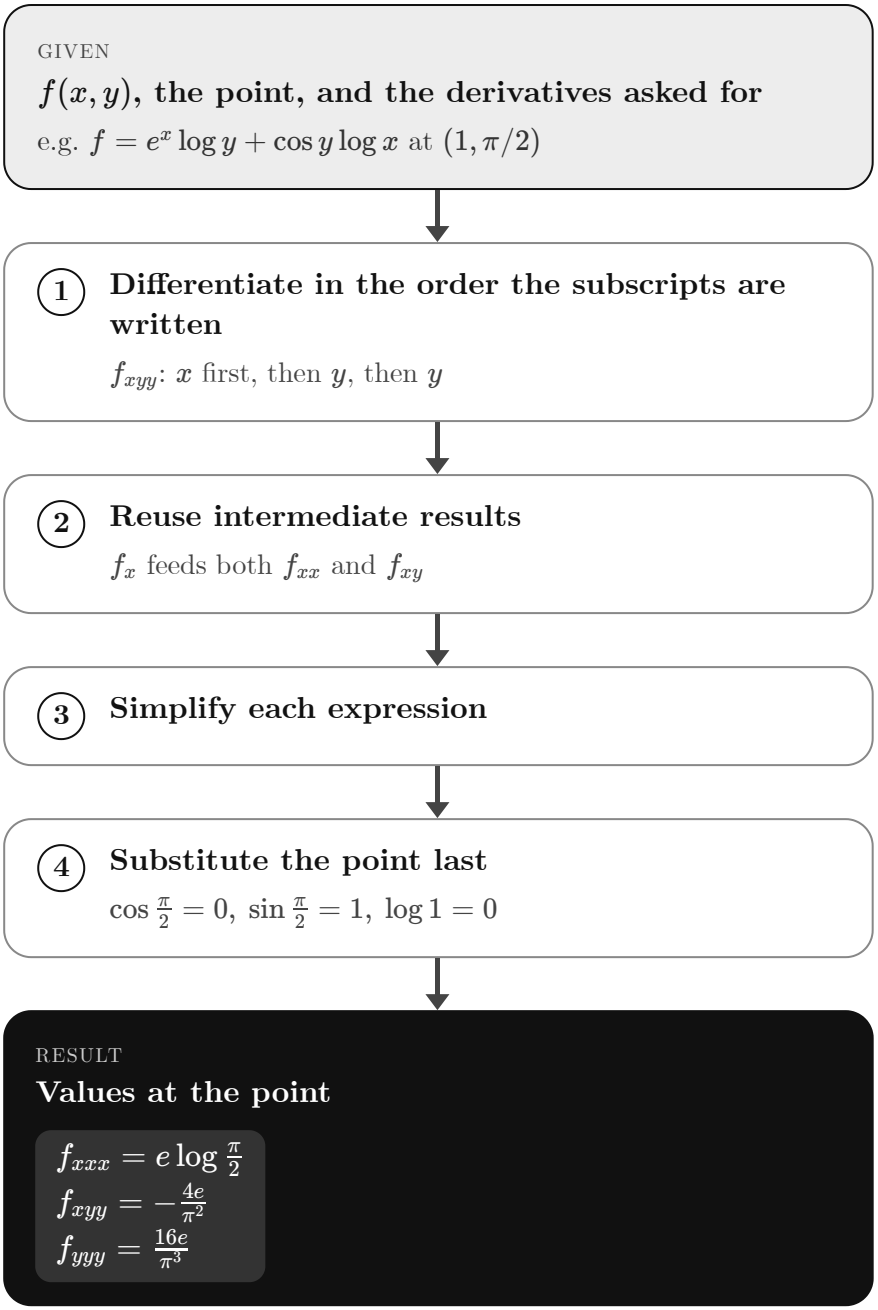
7 Problems on partial derivatives



Exam trap. In x, x^y is a power; in y, it is an exponential. Using the wrong rule is the usual slip.

Figure F5. Verifying equal mixed partial derivatives. ESA ★★

CW U1 P2 · T002 00:14:59



Remark. log is the natural logarithm, so $\frac{d}{dy}\log y = \frac{1}{y}$. Check one value against the CW answer key if you can.

Figure F7. Evaluating higher-order partials at a point (ESA Dec 2023 Q1.a). ESA ★★

ESA Dec-23 1a · CW U1 P5 · S002·3

· One variable vs two variables

	One variable, $y = f(x)$	**Two variables, $z = f(x, y)$**
Graph	Curve in a plane	Surface in 3-D
Tangents at a point	One	Infinitely many, one per direction
First derivative	dy/dx: slope of the tangent	f_x, f_y: slopes parallel to the x- and y-axes; ∇f for other directions
Second derivative	d^2y/dx^2: concave up or down	f_{xx}, f_{yy}: concavity in x, y; f_{xy}, f_{yx}: change of one slope along the other direction
Number of order n	1	2^n (or m^n for m variables)

ADDED FOR CLARITY S002·4 · S002·6 · S002·9 · S002·3